E-COMMERCE FULFILMENT & DELIVERY PERFORMANCE ANALYSIS

Business Problem Definition & Analytical Scope

Dataset: Olist Brazilian E-Commerce Public Dataset (Kaggle)

Analytical approach: Python, pandas, SQL/SQLite and Power BI

Project type: End-to-end operational analytics portfolio project

1. Business Context

An e-commerce fulfilment operation must coordinate order processing, sellers, shipment handover, customer delivery and post-delivery feedback. Late deliveries, freight cost variation and low customer review scores can signal service friction and operational inefficiency. Management needs a consistent view of delivery performance across customers, states, sellers and product categories.

2. Business Problem Statement

What factors are associated with late delivery, freight cost variation and customer review scores in an e-commerce fulfilment operation? The project uses historical Olist order, item, customer, seller, payment, product, review and geolocation data to describe fulfilment performance and identify areas that warrant operational investigation.

3. Project Objectives

• Measure delivery duration and late-delivery rate for delivered orders with valid actual and estimated delivery timestamps.

• Assess how review scores differ between late and on-time/early delivered orders.

• Compare delivery performance across customer states, using order counts alongside rates.

• Analyse freight value and delivery exposure by product category and seller geography.

• Create a reproducible data-quality, cleaning, feature-engineering and EDA workflow.

• Prepare nine cleaned relational CSV tables for Power BI data modelling.

4. Key Business Questions

• What percentage of eligible delivered orders arrived after the estimated delivery date?

• What are the average and median delivery days from purchase to customer receipt?

• Are late deliveries associated with lower customer review scores?

• Which customer states have higher late-delivery rates, and how many eligible orders support each rate?

• Which product categories show higher freight value or late-delivery exposure?

• Where is seller activity concentrated, and how does it relate to item/order volume?

• Which operational areas should be prioritised for further investigation?

5. KPI Definitions

KPI	Definition
Eligible delivered orders	Delivered orders with valid purchase, actual delivery and estimated delivery timestamps.
Late delivery rate	Late eligible delivered orders ÷ all eligible delivered orders.
On-time delivery rate	Eligible delivered orders delivered on or before estimated date ÷ eligible delivered orders.
Delivery days	Actual customer delivery timestamp minus purchase timestamp, expressed in days.
Delay days	Actual customer delivery timestamp minus estimated delivery timestamp, expressed in days.
Average review score	Mean review score for the selected order population; show review coverage.
Freight value	Freight amount from order-item records; report per item or aggregated per order with the grain stated.

6. Dataset Scope and Table Structure

Source table	Analytical role
Customers	Customer identifier and customer ZIP/city/state.
Geolocation	ZIP-prefix coordinate and location lookup; raw source may contain multiple rows per prefix.
Order items	Item-level product, seller, price, freight and shipping-limit information.
Payments	Payment sequence, type, instalments and payment value.
Reviews	Review score and review text/timestamps.
Orders	Order status and purchase, approval, carrier, actual delivery and estimated delivery timestamps.
Products	Product category and physical/product metadata.
Sellers	Seller identifier and seller ZIP/city/state.
Category translation	Portuguese-to-English product category mapping.

7. Methodology

The workflow covers data inventory and profiling; missing-value, duplicate and key checks; timestamp conversion; text/numeric standardisation; delivery feature engineering; order/item-grain analysis; descriptive EDA; and export of nine separate cleaned tables. Payments and reviews are aggregated to order grain before being joined to order-level analysis to avoid multiplying order records.

8. Data Modelling and Deliverables

The cleaned output will retain nine separate CSV tables: customers, geolocation, order_items, payments, reviews, orders, products, sellers and category_translation. Analysis joins are temporary and used only for the relevant EDA.

Power BI should preserve natural table grains rather than load one flattened all-purpose fact table. Geolocation will be aggregated to one record per ZIP prefix for use as a lookup dimension; the original Kaggle input remains unchanged.

9. Assumptions and Limitations

• Late delivery is measurable only for delivered orders with both actual and estimated delivery timestamps; other statuses remain in the cleaned orders table but are excluded from late-rate denominators.

• Missing delivery dates may be structurally expected for cancelled, unavailable or otherwise non-delivered orders; they must not be deleted indiscriminately.

• Review comments are optional text fields; missing comments do not necessarily mean a missing review score.

• Freight value currency convention must be confirmed before adding a currency symbol or comparing with external financial data.

• State and category rates must be accompanied by their supporting order/item counts.

• Descriptive associations do not establish causality. Operational root causes require further process-level evidence.

• The dataset is historical and does not support live tracking or an operational forecast without additional timestamped, prospective data.

10. Expected Business Use

The analysis is intended to help fulfilment and operations stakeholders identify where delivery performance varies, where freight costs differ, and whether customer feedback patterns align with delivery outcomes. The outputs support prioritisation of investigations and the design of controlled operational improvements; they do not claim guaranteed savings or causal effects.

Prepared as a project definition document. Dataset-derived findings and numerical results must be added only after the Kaggle notebook has been executed and its outputs validated.